AI NEXUS IT

COMPLETE PROGRAM SYLLABUS

Python | Data Science | Machine Learning | AI & Deep Learning
Django | SQL Server & MongoDB | Git & Docker
AWS & Microsoft Azure/Fabric | Power BI & Excel | PySpark | Linux

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Mode	Online & Offline Classes | 3 Real-Time Projects

Table of Contents

MODULE 1

Technology Stack Overview

Roles	Python Developer, ML Engineer, Data Scientist, Data Engineer, Data Analyst, BI Developer
Programming Language	Python (Basic → Advanced)
Web Frameworks	Django, Flask, FastAPI
Databases	SQL Server (T-SQL), MongoDB, PostgreSQL, MySQL
DevOps Tools	Git, Docker, Kubernetes, Ansible, Jenkins (CI/CD)
AWS Cloud	S3, EC2, Step Functions, Lambda, Glue, Redshift, EMR, CloudWatch, RDS, IAM, SNS, SQS, Sage
Microsoft Azure & Fabric	Azure Data Factory, Azure Synapse Analytics, Microsoft Fabric (Lakehouse, Data Engineering, Data
Machine Learning Stack	Pandas, NumPy, Seaborn, Matplotlib, Scikit-Learn, EDA, Statistics
ML Algorithms	Linear & Logistic Regression, Decision Tree, KNN, K-Means, Random Forest, XGBoost, SVM
Big Data Technology	PySpark, Hadoop
Visualization / BI Tools	Power BI, Tableau, Microsoft Excel (Pivot Tables, Power Query, Power Pivot, VBA basics)
Deep Learning	TensorFlow, Keras, CNN, RNN, LSTM, GRU
API / REST Framework	Django REST Framework, Flask API — creation & consumption
API Testing Tool	Postman
Front End	HTML, CSS, JavaScript, Bootstrap, PHP
Model Deployment	Nginx, Gunicorn, Flask/Django deployment, Cloud deployment
Operating Systems	Unix, Linux (CentOS/RedHat), Amazon Linux, EC2
Collaboration Tools	VPN, Jira, Slack, MS Teams, Confluence — real-time industry workflows
Projects	3 Real-Time, Industry-Style Projects across the program

Additions vs. the original syllabus: Microsoft Fabric, an expanded AWS service list, a dedicated Power BI module, and a dedicated Excel-for-Analytics module have been added (sections 17–19) since these were referenced only partially in the source document.

MODULE 2

Python — Basic Programming

1. Python Introduction — Part 1

- Industry orientation: team building approach, industry scenarios, market conditions, supply–demand opportunities
- Syllabus declaration and time management
- What is a language? Types of languages; Translators, Compiler, Interpreter, Debugger

2. Python Introduction — Part 2

- What is Python? Why Python? History and features of Python
- Why Python is a general-purpose / high-level language
- Limitations of Python

3. Python Installation & IDEs

- Python distributions, Anaconda Navigator
- Installation on Windows, Unix, Linux, Mac
- Online Python IDLE
- Real-time IDEs: Spyder, Jupyter Notebook, PyCharm; different modes of Python

4. Language Initials

- Python identifiers (rules & regulations)
- Basic data types (sequential, non-sequential, ordered, non-ordered)

5. List

- Definition, properties and types of list
- Indexing (forward & reverse); ordered/unordered, mutable/immutable nature
- List operations, methods and functions (Part 1 & 2)

6. Tuple

- Definition, properties and types of tuple
- Tuple operations, methods and functions

7. Set

- Definition and ways of creating a set; list vs. set
- Accessing set elements; set methods and operations; union of sets

8. Dictionary

- Definition; difference between list, set and dictionary
- Creating, accessing, copying and updating a dictionary
- Reading keys / values / items; deleting keys; sorting a dictionary
- Dictionary functions and methods

Mandatory Basic Skills

- VPN, Jira task management, Slack, Microsoft Teams, Confluence
- Real-time industry scenarios — client meetings and interaction etiquette

9. String

- Definition and properties of strings
- Processing via indexing and iterators; slicing and manipulation
- String operators and built-in methods

10. Type Casting

- String formatting and type casting

11. Operators

- Arithmetic, comparison, assignment, logical, bitwise, shift, membership and identity operators
- Ternary operator; operator precedence; difference between 'is' and '=='

12. Control Statements

- Conditional: if, if-else, if-elif-else ladder, nested if-else
- Loops: for, while, nested loops
- Branching: break, continue, pass
- Case studies — pattern making (letters and diagrams)

13. Functions

- Definition, advantages, syntax, calling/invoking functions
- Classification by argument/return value combinations
- Classification by parameter type: positional, default, variable-length (*args), keyword, **kwargs
- zip(); global vs. local variables

14. Anonymous Functions

- Lambda functions; map(), filter(), reduce()

15. Code Optimization

- List, tuple and dict comprehensions

16. Function Aliasing

- What is function aliasing and where it's used

17. Decorators

- What is a decorator, its uses and syntax
- Closures — inner functions; generators; iterators

18. Doubt-Solving Session

- Consolidation and open Q&A; on Basic Python

MODULE 3

Python — Advanced

19. Object-Oriented Programming

- Procedural vs. object-oriented programming
- OOP principles: encapsulation, abstraction (data hiding)
- Classes and objects; instance vs. class variables/methods; static methods
- 'self' and 'cls' reference variables; property() and getattr/setattr
- Inner classes (local & complex); accessing class-level and object-level members
- Access modifiers: private (__), protected (_), public
- Inheritance: single, multi-level, multiple, hierarchical, hybrid, diamond inheritance
- Method Resolution Order (MRO); super(); constructors in inheritance; Object class
- Duck typing (interview topic); abstract base classes
- Polymorphism: overriding (method & constructor) and overloading (method, constructor, operator)

20. Python Modules

- Modular programming, predefined vs. user-defined modules
- import, from...import, module aliasing; math and random modules

21. Packages

- Organizing a project into packages; package vs. folder
- PIP — installing/uninstalling packages

22. Exception Handling

- Syntax vs. runtime errors; common exception codes
- try–except, multi-except, finally; try-except-finally
- raise keyword; custom/user-defined exceptions; case studies

23. File & Directory Handling

- Opening files, file modes, reading/writing/appending data
- CSV module — creating, reading and writing CSV files

24. OS Module

- Shell commands via Python; creating/removing files & directories
- Renaming files/directories; executing system commands; shutdown/restart

25. Multi-threading & Multi-processing

- Multi-tasking vs. multi-threading; threading module
- Thread lifecycle; start(), sleep(), join(); single vs. multi-threaded apps

26. Object Serialization

- pickle module; XML parsing; JSON parsing

27. Python Logging

- Logging levels and configuration; overwrite mode; timestamps
- Logging exceptions; building a customized logger

28. Assertions

- Simple and augmented assertions; real-time use cases

29. Garbage Collection

- Manual garbage collection; self-referencing objects
- 'gc' module, collect() method, threshold function

30. Database Connectivity

- DBMS fundamentals; file system vs. DBMS
- Python–SQL Server / MySQL connectivity via connector modules
- cursor object methods: execute(), executemany(), fetchone(), fetchmany(), fetchall()
- Static vs. dynamic queries; transaction management
- Interview topic: Monkey patching

MODULE 4

Data Science with Python

NumPy

- Arrays and array processing
- Basic operations in NumPy
- Indexing and slicing

Pandas

- Series and DataFrames
- Indexing and slicing
- Groupby, concatenating, merging & joining
- Handling missing values
- Data input/output
- Pivot tables and cross-tabulation

Data Visualization — Matplotlib

- Line plots, histograms
- Box and violin plots
- Scatter plots, heatmaps, subplots

Data Visualization — Seaborn

- Statistical visualization with Seaborn

Working with Text — Regular Expressions

- Literals and meta-characters
- Using regular expressions with Pandas
- Inbuilt regex methods and pattern matching

Project — Data Mining

An end-to-end project that starts from scratch: collecting raw data from multiple sources and converting unstructured data into a structured format for Machine Learning and NLP models. Covers the four core stages of the Data Science life cycle — Data Collection, Data Mining, Data Pre-processing and Data Visualization (source formats: text, CSV, TSV, Excel files, matrices, images).

MODULE 5

Statistics — Descriptive & Inferential

Basic Terminology

- What is statistics and probability; role of statistics in data science
- Population vs. sample
- Sampling techniques: convenience, simple random, systematic, stratified, cluster

Variables & Data Types

- Dependent and independent variables
- Qualitative vs. quantitative data; categorical (nominal, ordinal); numerical (interval, ratio)
- Discrete vs. continuous data

Central Tendency & Dispersion

- Mean, median, mode
- Standard deviation and variance
- Box plot and distribution shape

Probability Fundamentals

- Probability vs. statistics; terminology and rules
- Marginal, joint, union and conditional probability

Probability Theory

- Conditional probability and Bayes' theorem
- Confusion matrix
- Z-score
- Histogram analysis

Probability Distributions

- Expectation, variance, skewness, kurtosis
- Discrete: Bernoulli, Binomial, Geometric, Poisson
- Continuous: Exponential, Normal/Gaussian, t-Distribution
- Confidence interval, standard error, margin of error

Statistical Testing

- Hypothesis testing
- Chi-square test
- t-test
- ANOVA

MODULE 6

Machine Learning — Supervised Learning

Introduction

- What is Machine Learning?
- Supervised vs. unsupervised learning
- Regression vs. classification models

Linear & Multiple Regression

- Regression fundamentals; hands-on use case
- Residual analysis; feature reduction (AIC), multicollinearity, influential points
- Hypothesis testing of the model; confidence interval of slope
- R-square and goodness of fit; leverage; polynomial regression; categorical variables

Logistic Regression

- Logistic regression intuition and the logit function
- Hands-on business case
- Evaluation: confusion matrix, accuracy, precision, recall, ROC curve

Naive Bayes Classifier

- Joint & conditional probability review
- Model assumptions and probability estimation
- Data processing, feature selection, classification

Principal Component Analysis (PCA)

- Why dimensionality reduction is needed
- Eigenvalues, eigenvectors, orthogonality
- Feature extraction; applications

Time Series Forecasting

- Trend, cyclical and seasonal analysis
- Smoothing, moving averages, auto-correlation, ARIMA
- Applications in financial markets

Decision Tree

- Decision & leaf nodes; variable selection; branching; stopping criteria
- Pruning and tree depth; overfitting
- Metrics: Gini impurity, information gain, variance reduction
- Regression trees; interpreting a tree via if-else; pros/cons; cross-validation

K-Nearest Neighbor (KNN)

- KNN algorithm and regression; curse of dimensionality
- Outlier treatment and anomaly detection; cross-validation; pros and cons

Support Vector Machine (SVM)

- Linear learning machines, kernel space
- Hands-on classification and regression using a business case

Ensemble Methods

- Bias–variance tradeoff; bagging & boosting
- Random Forest, Gradient Boosting, XGBoost

Case Studies

- Predictive analytics
- Banking — customer service prediction
- Healthcare — heart disease & diabetes prediction
- Insurance use cases
- Telecom churn prediction
- Bike sharing demand
- Air quality forecasting

MODULE 7

Machine Learning — Unsupervised Learning

Clustering

- Clustering methods and distance measures
- Iterative distance-based clustering
- Handling continuous and categorical values in K-Means
- Hierarchical and density-based clustering
- Cluster stability testing
- Hands-on implementation in Python

Recommendation Systems & Association Rules

- Combining clustering and classification
- Mathematical model for association analysis; Apriori algorithm
- Rule metrics: lift, support, confidence, conviction
- Collaborative filtering and content-based learning

MODULE 8

Natural Language Processing (Text Mining)

Introduction

- What is text mining?
- Libraries: NLTK, spaCy, TextBlob
- Structured vs. unstructured data
- Extracting text from files and websites

Text Preprocessing

- Regular expressions for pattern matching
- Text normalization
- Sentence & word tokenization; text segmentation
- Stemming and lemmatization

Natural Language Understanding

- Automatic and N-gram tagging; transformation-based tagging
- Bag of Words; POS tagging; TF-IDF; cosine similarity
- Vector space model; Named Entity Recognition; relation extraction

Matrix Factorization

- Singular Value Decomposition (SVD)

Text Indexing

- Inverted indexes; Boolean query processing
- Phrase and proximity queries; Latent Semantic Analysis

Text Classification

- Building and evaluating text classifiers

Case Studies

- Text mining
- Sentiment analysis
- Spam detection
- Dialogue prediction

MODULE 9

Artificial Intelligence & Deep Learning

Introduction to Neural Networks

- Perceptron; activation functions; cost functions
- Gradient descent; stochastic gradient descent; back-propagation

Deep Learning Frameworks

- Installing TensorFlow and Keras
- TensorFlow/Keras basic syntax
- TensorFlow graphs; variables & placeholders
- Saving/restoring models; TensorBoard

Artificial Neural Networks (ANN)

- ANN for regression and classification
- Evaluating, improving and tuning the ANN

Convolutional Neural Networks (CNN)

- Convolution operation, ReLU layer, pooling
- Flattening, full connection, softmax and cross-entropy

Building CNNs in Python

- Introduction to computer vision; OpenCV basics
- Image/video operations; geometric transformations (rotation, affine, perspective)
- Thresholding, contours, edge detection, morphological transformation
- Harris corner detection; reshaping & normalizing images
- Building and training a CNN for image classification

Keras (TensorFlow Backend)

- Keras vs. TensorFlow
- Building ANN and CNN with Keras

Sequential Models (NLP)

- Recurrent Neural Networks and the vanishing gradient problem
- LSTM (Long Short-Term Memory)
- GRU (Gated Recurrent Unit)

Projects

Face Recognition — collects and preprocesses images, applies an ML/DL model, trains and tests for gender and identity recognition. Use cases: security unlock, gender recognition, identity verification.

Chatbot — a virtual-assistant style project combining NLP, Deep Learning and AI to build an interactive, responsive conversational system (e.g., Alexa, Siri, Google Assistant style).

MODULE 10

Model Deployment

- Creating pickle and frozen model files
- Cloud deployment of Machine Learning and Deep Learning models for production
- Serving models with Flask/Django APIs behind Nginx and Gunicorn

MODULE 11

Django Web Framework (with Live Project)

Introduction to Front End

- HTML, CSS, JavaScript, Bootstrap

Introduction to Django

- Features of Django
- Django web server
- Django environment setup
- A simple 'Hello World' application

Displaying Hyperlinks — Project

- Django architecture; MVC and MTV patterns
- Starting a project and apps; activating the first app
- Views, URL mapping, templates and template inheritance
- Sending data: URL → view → template

Creating a Website — Project

- Creating an app inside the project; models
- Migrating models to tables; field examples
- Django shell — save, retrieve, modify, sort, filter and delete objects
- Modifying the data model

Administration Panel

- Using and customizing the admin interface
- Adding users; data access/modification via admin; permissions

Building the Site's First Page

- Django template system and inheritance
- Styling: background colours, banners, background images
- Storing/displaying images; user registration and uploads

Django Forms

- Forms basics; creating a 'Contact Us' form; form field examples

Django Email Functionality

- Configuring email settings; sending emails with Django

Django Template Language

- Template tags: if/elif/else, for, comments, filters
- Using templates to render site data

Integrating Bootstrap

- Tables, grids, carousels

Sessions & Cookies

- Difference between session and cookie
- Creating sessions/cookies in Django

Other Databases in Django

- SQLite; configuring and working with SQL Server / MySQL
- Configuring and working with Oracle

Django REST Framework

- Building CRUD operations via REST APIs

Live Project Implementation

- Full project life cycle; building a functional Django website

MODULE 12

SQL Server (Transact-SQL) — Database Programming

Relational database training built on Microsoft SQL Server and T-SQL.

Course Objectives

- Learn database models; overview of SSMS and T-SQL
- Write simple and complex queries; aggregate queries (SUM, MIN, MAX, AVG)
- Insert, update, delete and retrieve data; joins and sub-queries
- Work with data types, stored procedures, functions, views and triggers
- Design and maintain databases; manage transactions; retrieve data via cursors
- Manage binary data using BLOBs

Introduction to DBMS

- File management vs. DBMS; physical & logical data models
- Hierarchical, network, relational, object and object-relational models
- Conceptual model: Entity–Relationship (E-R)

Introduction to SQL Server

- SQL Server vs. Oracle vs. DB2
- Connecting to server; server type/name; authentication modes
- SQL Server Management Studio: Object Explorer, Query Editor

T-SQL Fundamentals

- History & features of T-SQL
- Command types: DDL, DML, DQL, DCL, TCL
- Creating/altering/deleting databases; constraints (Not Null, Unique, Default, Check, PK, FK)
- Data types; creating/altering/deleting tables

Data Manipulation Language

- Insert, Identity; creating a table from another table
- Update (with computed columns); Delete vs. Truncate

Data Query Language

- SELECT, WHERE, ORDER BY, DISTINCT, ISNULL(), column aliases
- Predicates: BETWEEN...AND, IN, LIKE, IS NULL

Built-in Functions

- Scalar: numeric, character, conversion, date functions
- Aggregate & statistical aggregate functions
- GROUP BY / HAVING; super aggregates; OVER(PARTITION BY…)
- Ranking functions; Common Table Expressions (CTE); TOP n clause

Set Operators & Joins

- UNION, INTERSECT, EXCEPT
- Inner, equi, natural, non-equi, self and outer joins (left/right/full); cross join

Sub-Queries

- Single-row and multi-row sub-queries; ANY/SOME, ALL
- Nested and correlated sub-queries; EXISTS / NOT EXISTS

Indexes

- Clustered and non-clustered indexes; create/alter/drop

Security

- Login creation (SQL & Windows authentication); user creation
- Granting/revoking permissions; roles

Views

- Purpose of views; create/alter/drop; simple vs. complex views
- Encryption and schema-binding options

Transaction Management

- BEGIN/COMMIT/ROLLBACK/SAVE transaction
- Role of the log file; implicit transactions

T-SQL Programming

- Control statements: IF, CASE; looping: WHILE
- Cursors: forward-only, scroll, static, dynamic, keyset, local, global

Stored Procedures & Functions

- Creating/altering/dropping stored procedures
- Optional, input and output parameters; permissions
- User-defined functions: scalar, table-valued (inline & multi-statement)

Triggers

- Purpose; differences vs. stored procedures/UDFs
- Create/alter/drop; magic tables; INSTEAD OF triggers

Exception Handling

- Implementing exception handling; custom error messages; RAISE

CLR Integration

- What CLR integration is and how to implement it

Additional Topics

- Working with the XML data type
- Backup & restore; attach & detach databases
- Normalization

MODULE 13

NoSQL Database — MongoDB

Overview

- Introduction to MongoDB and NoSQL
- Advantages over RDBMS
- MongoDB data types; installation; data modeling

Operators

- Query & projection operators
- Update operators
- Aggregation pipeline stages
- limit(), sort(), query modifiers

Database Commands

- Aggregation, geospatial, query/write operation commands
- Query plan cache, authentication, user & role management
- Replication and sharding commands; session commands

Database & Collections

- Create/drop database; create/drop collection

CRUD — Documents

- Insert, update, delete, query documents
- SQL-to-MongoDB mapping; text search; partial updates & document limits
- Multi-update, upsert, wire protocol, bulk() operations
- Common commands: runCommand(), isMaster(), serverStatus(), currentOp()/killOp(), collection.stats()/drop()

Shell, Cloud & Tools

- MongoDB shell collection/cursor methods
- MongoDB Stitch, Atlas, Cloud Manager, Ops Manager
- MongoDB Compass and BI Connector

Connectivity

- Java, PHP and Python drivers for MongoDB

MODULE 14

Git & Version Control

Introduction

- What is a Version Control System (VCS)?
- Distributed vs. non-distributed VCS
- What is Git; alternatives; GitHub/GitLab/Bitbucket

Installation & Key Terminology

- Installing and configuring Git; GUI tools
- Clone, working tree, checkout, staging area, add, commit, push, pull, stash

Local Repository Actions

- init, status, add, commit, reset, rm, log

Remote Repository Actions

- Creating a remote repo; clone, push, pull

Tagging & Branching

- Lightweight and annotated tags; listing, checking out, pushing/pulling tags
- Creating, listing, checking out, pushing and pulling branches; understanding HEAD

Merging & Workflows

- fetch, rebase, pull
- Centralised, Feature Branch, Gitflow and Forking workflows

Stashing

- Using stash; creating a branch from a stash

Advanced Actions

- git clean, reset, revert, checkout of a previous commit
- Fast-forward vs. three-way merge; resolving conflicts; cherry-pick

Advanced Configuration

- Aliases, submodules, patches, hooks

MODULE 15

Docker & Containerization

Module 1 — Docker World

- Introducing Docker; VM vs. Docker; Docker architecture; Docker Hub
- Installing/configuring docker-machine; containerizing and running images
- Port forwarding; exercises: install Docker, create images, expose container ports

Module 2 — Dockerfile, Builds & Networking

- Dockerfile directives: USER, RUN, ENV, CMD vs. RUN, ENTRYPOINT, EXPOSE
- Volume management basics; Docker networking: list/inspect/create/remove/assign
- Exercises: custom image from Dockerfile, container management, adding external content

Module 3 — Commands & Structures

- Inspecting container processes; controlling port exposure
- Naming containers; Docker events; managing/removing images
- Saving/loading images; image history and tagging; pushing to Docker Hub
- Basics of CI for Docker

Module 4 — Docker Compose

- Default network, isolating containers, aliases & container names, links
- Configuring and running Compose environments; bringing environments up/down

MODULE 16

AWS Cloud for Data Science & Data Engineering

Core AWS services used across Data Science, Data Engineering and Data Analyst workflows.

- Amazon S3 — object storage for data lakes
- Amazon EC2 — compute instances for training/serving
- AWS Lambda — serverless compute for pipelines and triggers
- AWS Step Functions — orchestrating multi-step workflows
- IAM — identity, access management, roles and policies
- Amazon SNS — pub/sub notification service
- Amazon SQS — managed message queues
- Amazon SageMaker — build, train and deploy ML models
- Amazon ECR — container image registry
- Amazon SES — transactional email service
- Amazon RDS — managed relational databases
- Amazon Redshift — cloud data warehousing
- Amazon DynamoDB — managed NoSQL database
- AWS Glue — serverless ETL and data cataloging
- Amazon Kinesis — real-time streaming data
- Amazon Athena — serverless SQL querying over S3
- Amazon QuickSight — cloud-native BI dashboards
- Amazon CloudWatch — monitoring, logging and alarms
- Amazon ECS — container orchestration
- Amazon EMR — managed Hadoop/Spark clusters
- AWS CodeCommit / Git integration — source control on AWS
- AWS CloudFormation — Infrastructure as Code (added for end-to-end pipeline deployment)
- Amazon EventBridge — event bus for decoupled, event-driven pipelines (added)

MODULE 17

Microsoft Azure & Microsoft Fabric

New module added to complete the cloud & analytics stack.

Microsoft Azure Fundamentals

- Azure account, subscriptions, resource groups and regions
- Azure Resource Manager (ARM); role-based access control (RBAC)
- Azure Data Lake Storage (Gen2) — hierarchical data lake storage
- Azure Logic Apps — workflow automation and integration
- Azure ML Studio — building and deploying ML models on Azure
- Azure Data Factory (ADF) — pipelines, datasets, linked services, triggers, data flows
- Azure Synapse Analytics — unified data warehousing and big data analytics

Microsoft Fabric — Overview

- What is Microsoft Fabric; the SaaS, unified analytics platform concept
- OneLake — a single, unified data lake for the whole organization
- Fabric workspaces, capacities and licensing basics
- Fabric experiences: Data Engineering, Data Factory, Data Science, Data Warehouse, Real-Time Intelligence, and Power BI

Fabric — Data Engineering & Lakehouse

- Creating a Lakehouse; Delta tables and file management
- Notebooks (PySpark/Python) for transformation inside Fabric
- Fabric Data Factory — pipelines and Dataflows Gen2 for ETL/ELT
- Shortcuts — referencing data across OneLake without duplication

Fabric — Data Warehouse & Real-Time Intelligence

- Fabric Data Warehouse — T-SQL based warehousing over OneLake
- Real-Time Intelligence — event streams, KQL databases and real-time dashboards

Fabric — Power BI Integration

- Building semantic models (datasets) directly on Lakehouse/Warehouse data
- DirectLake mode — high-performance reporting without data duplication
- Publishing and sharing Power BI reports inside a Fabric workspace

MODULE 18

Power BI — Business Intelligence & Visualization

New, fully detailed module (previously listed only as a tool name).

Introduction to Power BI

- What is Power BI; Power BI Desktop, Service and Mobile
- Power BI architecture; installing and licensing overview (Free, Pro, PPU, Premium)

Connecting & Transforming Data

- Connecting to Excel, CSV, SQL Server, Web, Folder and cloud sources
- Power Query Editor — shaping and cleaning data
- Merge queries, append queries, pivot/unpivot columns
- Data types, conditional columns, custom columns (M language basics)

Data Modeling

- Star schema vs. snowflake schema
- Relationships — cardinality and cross-filter direction
- Managing multiple tables; hiding fields from report view

DAX (Data Analysis Expressions)

- Calculated columns vs. measures
- Aggregation functions: SUM, AVERAGE, COUNT, DISTINCTCOUNT
- Time intelligence: TOTALYTD, SAMEPERIODLASTYEAR, DATEADD
- CALCULATE, FILTER, ALL, related functions; variables in DAX

Building Reports & Visuals

- Bar, line, pie, area, combo charts; tables and matrices
- Slicers, filters (visual/page/report level); drill-down and drill-through
- Bookmarks and buttons for interactivity; tooltips

Dashboards & Sharing

- Publishing reports to Power BI Service
- Creating dashboards; pinning tiles; sharing with workspaces and apps
- Row-Level Security (RLS); scheduled data refresh; Power BI Gateway

Advanced Topics

- Power BI and Microsoft Fabric integration (DirectLake)
- Performance optimization tips; Power BI with Excel and SQL Server as sources

MODULE 19

Microsoft Excel for Data Analytics

New module added to strengthen the Data Analyst track.

Excel Fundamentals

- Workbook/worksheet structure; cell referencing (relative, absolute, mixed)
- Data entry, formatting, conditional formatting, data validation

Formulas & Functions

- Logical: IF, AND, OR, IFS, nested IF
- Lookup: VLOOKUP, HLOOKUP, INDEX-MATCH, XLOOKUP
- Text functions: LEFT, RIGHT, MID, TRIM, CONCAT/TEXTJOIN
- Date/time functions; SUMIF/SUMIFS, COUNTIF/COUNTIFS, AVERAGEIF/AVERAGEIFS

Data Analysis Tools

- Sorting and filtering; removing duplicates
- PivotTables and PivotCharts — summarizing large datasets
- What-If Analysis: Goal Seek, Data Tables, Scenario Manager

Power Tools in Excel

- Power Query in Excel — importing and transforming data
- Power Pivot — data modeling and DAX measures inside Excel
- Creating relationships across multiple tables in the Excel Data Model

Visualization in Excel

- Charts: bar, line, pie, combo, sparklines
- Dashboard building using PivotCharts, slicers and form controls

Automation Basics

- Introduction to Macros and VBA basics for repetitive tasks
- Recording a macro; simple VBA scripting for automation

Excel with Python/SQL

- Reading/writing Excel files with pandas (openpyxl)
- Connecting Excel to SQL Server as a data source

MODULE 20

PySpark & Big Data Processing

Introduction

- What is Apache Spark? Why PySpark; Spark Python vs. Scala
- PySpark features; real-life usage
- SparkSession, SparkContext, RDD, parallelize()
- repartition() vs. coalesce(); broadcast variables; accumulators

RDD Computation

- Operations on an RDD; Direct Acyclic Graph (DAG)
- RDD actions and transformations; RDD persistence and its features
- Persistence options: MEMORY_ONLY, MEMORY_SER_ONLY, DISK_ONLY, DISK_SER_ONLY, MEMORY_AND_DISK_ONLY

Core Computing

- Fault-tolerance model in Spark; ways of creating an RDD; word count example
- Increasing partitions; aggregations over structured data with reduceByKey()

Groupings & Aggregations

- Single/multiple grouping with single/multiple aggregation
- reduceByKey() vs. groupByKey(); reduce() function; built-in transformations

Actions & Transformations

- countByKey(), countByValue(), sortByKey(), zip(), union(), distinct()
- Joins (inner/outer), cartesian(), cogroup() and other actions

PySpark SQL — DataFrame

- SQLContext; DataFrame & Dataset APIs; RDD vs. DataFrame vs. Dataset
- Creating DataFrames; converting RDD ↔ DataFrame ↔ Pandas
- StructType/StructField, Row and Column classes
- select(), collect(), withColumn(), where()/filter(), drop()/dropDuplicates()
- orderBy()/sort(), groupBy(), join(), union()/unionAll()/unionByName()
- UDFs, map(), flatMap(), foreach(); sample()/sampleBy(); fillna()/fill()
- pivot(), partitionBy(); ArrayType and MapType columns

PySpark SQL & Built-in Functions

- Aggregate, window, date/timestamp and JSON functions
- when(), expr(), lit(), split(), concat_ws(), substring(), regexp_replace()
- to_timestamp(), to_date(), date_format(), datediff(), months_between()
- explode(), array_contains(), collect_list()/collect_set(), struct()
- Ranking: row_number(), rank(), dense_rank(), percent_rank()
- JSON: from_json(), to_json(), json_tuple(), get_json_object(), schema_of_json()

External Sources & Deployment

- Spark–Hive and Spark–MySQL/SQL Server integration
- Working with CSV/JSON; narrow vs. wide transformations

- Adding/dropping/renaming columns; handling nulls; joins; window functions
- Writing to external sources; internal & temporary tables
- Deployment modes: Local, Standalone, YARN cluster
- Stages, tasks, driver/executor; building and tuning Spark pipelines
- PySpark Streaming concepts; Kafka integration; PySpark MLlib

MODULE 21

Linux / Unix Administration & Shell Scripting

Module 1 — Linux Concepts

- What is Linux; everyday use; Unix vs. Linux

Module 2 — Download, Install & Configure

- Oracle VirtualBox setup; creating a virtual machine
- Linux distributions; installing CentOS/RedHat; Linux Desktop (GUI); Linux vs. Windows

Module 3 — System Access & File System

- Accessing Linux via Putty; ifconfig/ip commands
- File system structure, navigation, paths; creating files/directories; file types
- find vs. locate; changing password; wildcards; soft/hard links (ln)

Module 4 — Linux Fundamentals

- Command syntax; file permissions (chmod); ownership (chown, chgrp)
- man/whatis; tab completion; tee; pipes
- Text processing: cut, sort, grep, awk, uniq, wc; diff/cmp; tar/gzip/gunzip
- truncate; cat and split; Linux vs. Windows commands

Module 5 — System Administration

- vi editor; sed command; user account management
- su and sudo; monitoring users (users, wall, write)
- System utilities: date, uptime, hostname, which, cal, bc
- Process/schedule management: systemctl, ps, top, kill, crontab, at
- Monitoring: df, dmesg, iostat, netstat, free; shutdown/reboot/halt/init
- System logs (/var/log); hostnamectl; uname; arch; recovering root password

Module 6 — Shell Scripting

- Linux kernel and shell types; writing basic shell scripts
- If-then, for-loop, do-while and case-statement scripts; aliases and command history

Module 7 — Networking, Servers & Updates

- Enabling networking in a Linux VM; ping, ifconfig, netstat, tcpdump
- ethtool, NIC/port bonding; wget, curl; ftp/scp file transfer
- yum/rpm package management; patch management; SSH/Telnet; DNS (nslookup, dig)
- NTP/chronyd; sendmail; Apache HTTP server; rsyslogd; OS hardening; OpenLDAP; traceroute

Module 8 — Disk Management & Run Levels

- System run levels; Linux boot process; Message of the Day
- Disk partitioning (df, fdisk); LVM configuration and extension; swap space; RAID
- File system check (fsck, xfs_repair); backup (dd); NFS

MODULE 22

Real-Time Projects & Career Toolkit

Real-Time Projects (3, across the program)

- Data Mining — end-to-end raw-data collection, cleaning, structuring, and visualization pipeline
- Face Recognition — image collection, preprocessing, ML/DL model training and testing for identity/gender recognition
- Chatbot — an NLP + Deep Learning + AI powered virtual assistant

Career & Workplace Readiness

- Version control and collaboration: Git, GitHub/GitLab/Bitbucket workflows
- Agile tools: Jira task management, Confluence documentation
- Team communication: Slack, Microsoft Teams, VPN access for remote work
- Client interaction and real-time industry meeting scenarios
- Interview preparation topics flagged throughout: MRO, duck typing, abstraction vs. encapsulation, monkey patching